



Coastal Marine Survey, LLC

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M/V “*BLACK JACK*”

REPORT OF MARINE SURVEY

OF THE VESSEL

“Black Jack”

**2002 Sea Ray
340 Amberjack**

CONDUCTED BY:
Scott M. Austin AMS®

Prepared for:
Mr. Jeff Davis

34SeaRay02
July 27, 2020

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I. INTRODUCTION

SCOPE OF SURVEY

At the request of the vessel buyer, **Mr. Jeff Davis**, the undersigned surveyor did complete a pre-purchase condition and valuation inspection of the M/V '**Black Jack**', a **2002 Sea Ray 340 Amberjack Open Express Cruiser**, on **July 27, 2020**. The vessel was found **in the water at berth #6, Islamorada Yacht Basin Lorelei Restaurant, and in Travellift slings at Coral Bay Marina, Islamorada, FL**. The vessel's Hull Identification Number (**SERF8938J102**) was verified from the starboard transom, and matches the hull number listed on the current USCG documentation paperwork, which was found onboard and appears to be in order. A basic sea-trial was performed, with results included throughout this report. The reason for the survey was to ascertain the physical condition and value of the vessel at the time of inspection. Ship's **battery DC power**, and **50A/250V split to two 30A/125V shore power cords** were used to check operation of the vessel's electrical systems.

No evaluation of the internal condition of the vessel's inboard gasoline engines, generator, and their components, or operating capacity of the vessel's propulsion systems was performed. For additional details on the internal condition, compression, and maintenance history of these engines and generator, I recommend consulting a reputable engine technician. Electronic equipment was checked for proper "power up" and basic function at sea-trial.

Unless specifically stated otherwise in this report, the surveyor visually inspected this vessel without removal of any fasteners, fixed or semi-fixed structures or equipment, and has not disassembled any hull structures, electronics, instruments, or machinery for inspection or testing.

CONDUCT OF SURVEY:

THE MANDATORY STANDARDS PROMULGATED BY THE UNITED STATES COAST GUARD (USCG), UNDER THE AUTHORITY OF TITLE 46 UNITED STATES CODE (USC), TITLE 33 AND 46, CODE OF FEDERAL REGULATIONS (CFR), AND THE VOLUNTARY STANDARDS AND RECOMMENDED PRACTICES DEVELOPED BY THE AMERICAN BOAT AND YACHT COUNCIL (ABYC) AND THE NATIONAL FIRE PROTECTION ASSOCIATION (NFPA) HAVE BEEN USED ONLY AS GUIDELINES IN THE CONDUCT OF THIS SURVEY. COMPLETE COMPLIANCE WITH SUCH STANDARDS VARIES WITH THE INTENDED SERVICE OF THE VESSEL, AND IS NOT GUARANTEED.

Use of **asterisks *** in the body of this report will indicate that a finding will be listed in the ***Findings and Recommendations*** section pertaining to the asterisked item. The findings reported herein reflect the conditions observed **at the time the survey was conducted**. Images supplied with this report were produced with a Nikon D-80 digital camera and accurately represent the vessel at the time of survey.

NOTE: **FP** (functions properly) **DNF/P** (does not function/properly) **AS** (appears sound) **ASWA** (appears sound where accessible) **F/GC** (fair/good condition)

II. GENERAL INFORMATION

TYPE OF SURVEY:	Pre-Purchase Condition and Value for Buyer	
NAME OF VESSEL:	“Black Jack“	
YEAR / MAKE / MODEL:	2002 / Sea Ray / 340 Amberjack Open Express	
YEAR BUILT:	2001	
BUILDER / BUILT:	Sea Ray Boats; Merritt Island, FL	
OVERALL VESSEL RATING:	**	Average (BUC)
EST. MARKET VALUE:	**	\$67,800.00 USD
EST. REPLACEMENT VALUE:	**	\$455,000.00 USD
HULL ID NUMBER (HIN):	SERF8938J102	
ALTERNATE HIN:	340AJ-523	
OFFICIAL NUMBER:	>> 1133062	expires 09/2020
STATE REGISTRATION NO:	TX# TX 8963 AN	expired 03/2020
HAILING PORT:	Seabrook, TX	
LENGTH OVERALL (LOA):	* 38’ 0”	
LENGTH ON DECK:	* 33’ 6”	
BEAM:	* 13’ 5”	
DRAFT:	* 3’ 0” (minimum)	
WEIGHT / DISPLACEMENT:	* 16,500# (dry)	
OWNERS NAME:	Mr. Donald Anschutz	
DATE OF SURVEY:	July 27, 2020	
HULL TYPE:	Planing monohull; modified deep-Vee	
HULL MATERIAL:	Fiberglass Reinforced Plastic (FRP)	
PROPULSION SYSTEM:	Twin MerCruiser 370HP inboard engines / direct drive	
GENERATOR:	Westerbeke 4.5kW	
FUEL TYPE:	Gasoline	
FUEL CAPACITY:	* 276 US gallons (labeled)	
FRESH WATER CAPACITY:	* 50 US gallons	
HOLDING CAPACITY:	* 27.5 US gallons (labeled)	
AC POWER:	120V; two 30A/125V shore power; AC generator	
DC POWER:	12V; ship’s batteries	
INTENDED USE:	Recreational; inshore cruising; Lake Erie, PA	

Asterisks * in this General Information section refer to the source of such information as follows:

* Per manufacturer’s specifications

** Refer to Summary and Valuation Section

>> **The USCG documentation number is not found fixed within the vessel as required**

VESSEL DESCRIPTION:

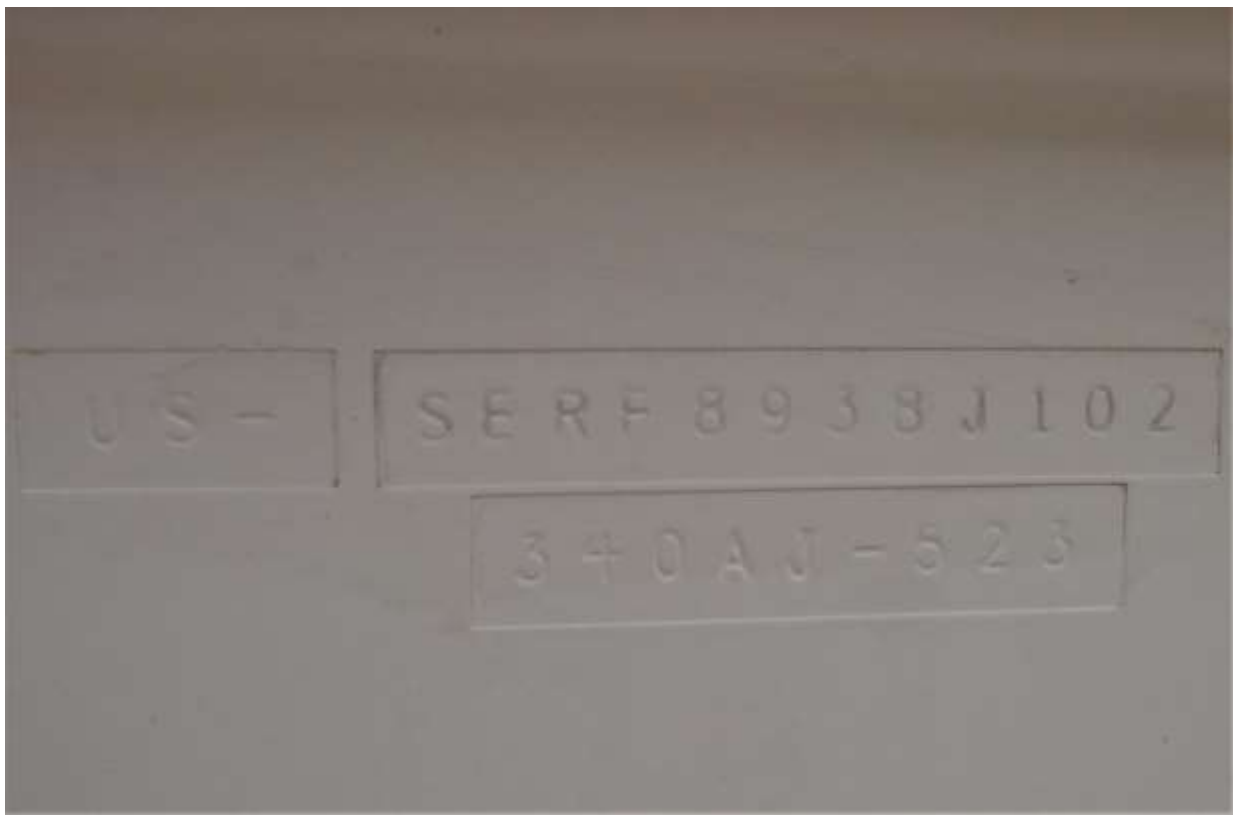
This 2002 Sea Ray 340 was found to be a fiberglass hull, twin gasoline inboard engine with generator, single helm station, aft cockpit express cruiser. The deep-Vee hull has hard chines, molded partial outer/mid/inner strakes, molded sheer-line spray-rails, drive pockets, and transom trim tabs. Forward at the bow is an integral anchor platform with a remote searchlight, and a stainless flume, with a sand anchor on two rollers, followed by three deck cleats, a starboard locker hatch, and a windlass with deck switches. Also at the bow is a fixed single stainless pulpit handrail on stainless stanchions, running continuously aft seventy-percent port and starboard. Continuing aft is a slightly raised open deck, with one deck hatch, followed by the cabin superstructure. Walkways lead aft around the cabin, with a wrap-around vented glass/aluminum frame windshield with two wipers, a molded navigation arch; fore/aft stainless frame/tan canvas awnings, and two portholes, then down two steps into the open aft cockpit. Forward in the cockpit, and up one step is the electric-lift bridge deck, with open dash and a center cabin companionway, followed to port by a transverse convertible leaning post/bench seat module, with a footrest and locker below, and outboard locker with AC breaker switches, while to starboard is the helm module, with center steering, full dash electronics, and dual engine controls, and a helm bench seat module, with ice-maker locker below. Back in the cockpit to port is a bench seat module, with icebox below, while to starboard is a covered prep station, wet bar, trash chute, and storage locker module. Moving aft is a port deck live well, and center fish box insert, followed by the enclosed transom, with gated port door to a fixed molded swim platform, with hide-away center hatch telescoping boarding ladder. The hull sides and transom are white gel coat, with a black/gold boot top/accent stripes, black/gold cabin stripes, and red/black/yellow transom graphics and lettering. The superstructure, decks, and cockpit are white gel coat, with white textured non-skid overlay in all traffic areas, and black anti-foul paint below the waterline. The vessel has a full-perimeter white PVC rub-rail with a stainless insert. The molded radar arch supports an open radar scanner, a satellite TV dome, a ratchet mount VHF antenna, a hailer speaker, and a GPS antenna. Also found are four cockpit cap-rail and six fixed coaming rod holders, stainless grab-rails at the bridge and cockpit sides, companionways, cabin sides, and transom, and three-side cockpit bolsters.

VESSEL ACCOMMODATIONS:

Forward below, aft of the mirrored chain locker bulkhead and reading lights, is a forward cabin. Forward in the cabin is a modified offset port queen berth, with locker drawer storage below, and outboard cut-out lockers, followed to port by a pivot bulkhead mount flat-screen TV, while to starboard is a full mirrored door hanging locker, with lower and outboard cabinetry. Moving aft out of the cabin, with sliding pocket triple door enclosure, enters the salon. Forward to port is a linear galley module, with gloss hardwood veneer cabinetry, bulkhead TV, sliding door cut-outs, beige Corian® countertop with a covered corner sink, two-element glass stovetop, upright refrigerator/freezer, and a cabinet microwave, followed by an enclosed head, with a molded round sink vanity, upper and lower cabinetry, a raised porcelain marine toilet, and a shower curtain and track. Starboard in the salon is outboard hardwood cabinetry, and a stepped fore and aft bench seating dinette module, with pedestal table, equipment access, and storage below, followed by stereo and the AC/DC electrical panel lockers. Aft in the salon is a three-step hinged companionway to the bridge deck and cockpit, with open storage and the main DC battery switch

ACCOMMODATIONS: (CONT.)

panel below. Storage is found in the chain locker, below the cabin berths, in all salon, cabin, galley, and head cabinetry and drawers, below cockpit and salon seating, in deck and sole hatches, in the deck box and live well, bridge lockers and wet bar/prep station module, below gunwale cut-outs, and in the aft bilge/steerage.



Transom Hull Identification Numbers

III. SYSTEMS

NOTE: USE OF *:

Use of * in the body of this report will indicate that a finding will be listed in the *Findings and Recommendations* section pertaining to the * item.

A. GALLEY:

Ventilation:	Companionway door, deck hatch, 12V exhaust fan vent – satisfactory
Stove Top:	Kenyon two-element, 120V; glass top; rails – GC, FP; both > 400°F
Safety:	‘On/Hot’ lights – FP
Oven:	N/A; (see Microwave below)
Refrigeration:	Norcold (m#DE441), 12/120V, upright/counter size – F/GC, FP
Latch/Seal:	Factory side rail/catch – FP / factory molded neoprene – GC
Secured:	Cabinet retainer – secure
Light:	Yes – FP
Freezer:	Top compartment (see above), plastic door – GC, FP; door stays open
Cockpit Ice-Maker:*	ULine (m# ULN-SP18WH-03), 120V, below helm seat – AS; DNF
Sinks:	Single; molded Corian corner/rectangle, medium well; cover insert – GC
Faucet:	Chrome two-knob hot/‘cold’ – GC, both FP; mild chrome pitting
Drain:	Overboard, port hull fitting, Marelon seacock – ASWA, handle FP
Hoses/Fittings:	Reinforced plastic, single clamps; nylon drain pipe/fittings – GC; no leaks
Microwave:	GE Turntable (m# JE520BW/005) carousel/plate, 110V, cabinet – GC, FP
Appliances:	N/A
Additional:	Assorted plastic galley utensils, dishes, glasses – all GC; not inventoried

COMMENTS AND OBSERVATIONS:

The port side galley is a framed linear module, with gloss hardwood veneer lower cabinetry, sliding door cut-outs above, wood shelving and accents, a white vinyl panel headliner, and tan carpet over laminate and wood sole. There is a beige Corian® countertop, with a covered, molded corner sink, two-element glass stovetop, upright refrigerator/freezer, and a cabinet microwave. Headliner and counter valance lighting, a 12V exhaust fan and cabin deck hatch ventilation, and two 110V dual outlets are also noted.

NOTE: The cockpit ice-maker appears sound, but does not function at the time of inspection, and the door latch is broken away. Service and prove the ice-maker, and replace the door latch.

B. SANITATION SYSTEMS:

Type:	Vacuum/freshwater; holding tank; overboard/pump-out; Type III MSD
Toilet:	SeaLand VacuFlush, white porcelain, hinged molded cover – GC, FP
Seat/Cover:	Molded plastic – GC; secure
Mounting:	Fasteners to raised sole – AS/secure
Vacuum Tank:	SeaLand VacuFlush, molded PVC pressure cylinder – ASWA, FP
Vacuum Pump:	SeaLand, V/G12VDC ‘duck valve’ vacuum generator – ASWA, FP
Controls:	Foot lever fresh water fill/flush – GC, FP

SANITATION SYSTEMS: (CONT.)

Holding Tank:	One; black water; with one vacuum generator tank (see above)
Construction:	Polyethylene; molded box – ASWA; no leaks noted
Capacity:	27.5 US gallons (labeled)
Location:	Port engine room, outboard
Secured:	Molded flanges/fasteners to sub-floor – ASWA; secure
Level Indicator:	Light indicator panel; toggle at salon panel – AS; near ½ at survey
Tank Vent:	Conventional; port hull fitting – ASWA
Filter:	Yes; SeaLand SaniGard inline canister at tank – AS; change routinely
Y-Valve:	N/A; direct to holding tank
Deck Fitting:	Stainless, port deck amidships, labeled ‘Waste’ – GC
Cap/Seal:	Stainless, O-ring, no tether – GC
Through-Hulls:	
Inlet:	N/A; freshwater flush (see Fresh Water)
Discharge:	Bronze seacock – AS; handle stiff; secure handle closed
Siphon Break:	ABS vented loop inline at discharge hose – ASWA; no leaks
Hoses/Fittings:*	Neoprene, single/double clamps – neoprene hoses aged/cracked
Macerator:	SeaLand/Leeson, T/W12VDC – powers up/AS; not proved overboard
Hoses/Fittings:*	Neoprene, single/double clamps – ASWA; leaks at discharge seacock
Sink:	Molded round, set in counter – GC
Faucet:	Grohe chrome retractable, 2-knob hot/’cold’ – GC, hot/’cold’ FP
Drain:	Overboard, starboard hull amidships, Marelson seacock – ASWA, FP
Hoses/Fittings:	Reinforced plastic, single clamps – ASWA, FP; no leaks noted
Shower:	Curtain/track open shower, bulkhead shower head bracket – GC, FP
Faucet:	Retractable at sink faucet (see above) – GC, FP; clean spray head
Drain:	Through sole drain, to Sump Box – ASWA; (see Sump Box)
Hoses/Fittings:	Reinforced plastic; single clamps at box – ASWA, no leaks noted
Sump Box:	Molded box, clear acrylic top, drains shower; A/C pan – F/GC
Pump/Switch:	Rule 500 bilge pump, Rule auto switch; salon breaker – AS, not proved
Hoses/Fittings:	Reinforced plastic, single clamps – ASWA

COMMENTS AND OBSERVATIONS:

The enclosed port head is a molded liner module, with molded cabinetry, a hardwood locking door, a molded sink vanity in a beige Corian countertop, vacuum flush porcelain toilet, overhead shower curtain and track, with a retractable wand at the sink faucet. The compartment headliner is molded liner, with textured non-skid sole. The head has a hinged mirror and cabinet doors, a 12V exhaust fan, overhead and valance lighting, a 110V dual outlet, towel and TP hardware, and an opening porthole. The sump box pump and auto-switch appear sound, but were not proved to pump overboard.

NOTE: The neoprene hose from the macerator pump to the below waterline discharge valve, is aged, cracked throughout, and leaks when the discharge valve is opened. Properly replace the head system discharge neoprene hose as necessary. The neoprene hose from the vacuum tank to the holding tank is aged and cracking. Replace the holding tank inlet neoprene hose as needed.

C. FRESH WATER SYSTEM:

Storage Tank(s):	One
Construction:	Polyethylene, molded box – ASWA
Capacity:	50 US gallons (reported by Sea Ray Boats)
Accessibility:	Limited top sight only (see Comments)
Location:	Centerline amidships; set linear below salon sole
Secured:	Aluminum bracket retainers, set over bilge – ASWA
Level Indicator:	Light indicator panel; toggle at salon panel – AS; at ½ at survey
Tank Vent:	Conventional; starboard amidships fitting – ASWA; not proved
Pressure Systems:	Manual; N/A
DC Electric:	Jabsco ParMax2.9 (m# 31395-0092), 12V diaphragm – GC, FP
Strainer:	N/A
Hoses/Fittings:	Water tubing/blue flex-hose; compression/swaged stainless ends – ASWA
Accumulator:	N/A
Water Heater:	Seaward (m# F-700), 115V – ASWA, FP at galley/head/transom outlets
Construction:	Stainless box cover, alloy element (reported) – ASWA
Capacity:	6 US gallons (labeled)
Location:	Port forward engine space
Secured:	Welded brackets/fasteners to sub-floor – ASWA; secure
Relief Valve:	Yes; plumbed overboard, Marelon hull seacock – AS/GC; not proved
Dock System:	Yes, port cockpit inlet hose bib – ASWA; FP at inspection
Regulator:	Reported; not sighted/not accessible
Deck Fill Fitting:	Stainless, starboard deck amidships, labeled ‘Water’ – GC
Cap/Seal:	Stainless, O-ring seal; chain tether – GC
Fill Lines:	Reinforced plastic hose, double clamps at tank – ASWA
Supply Lines:	Blue/red tubing, compression ends/fittings – ASWA; no leaks in system
Water Maker:	N/A

COMMENTS AND OBSERVATIONS:

The pressurized and dock supply freshwater systems function properly at the time of inspection. The system supplies pressurized tank water to the galley sink, head sink faucet, toilet, transom retractable shower wand, engine room, port cockpit, and bow locker hose bibs, prep sink faucet, ice-maker, and the water heater. Limited access to the water tankage prohibits a thorough inspection: the system appears sound where accessible at the time of inspection, with no leaks noted. (see **Hull/Decks section**)

NOTE: The wash down hose in the bow locker is in poor condition. Replace the bow locker hose as needed.

D. SALT/RAW WATER SYSTEMS:

Wash Down Pump:	ShurFlo ProBlaster (m# 2088-732-234), 12V diaphragm – AS, not proved
Inlet:	Bronze seacock – GC, handle FP
Strainer:	Perko bronze/acrylic – AS/GC; no leaks noted
Outlets:*	Starboard aft hose bib – handle frozen/DNF; system not proved
Hoses/Fittings:	Neoprene, double clamps – ASWA, GC
Live Well:	Molded port cockpit deck liner box – GC
Inlet:	Bronze seacock – GC, handle FP
Strainer:	Perko bronze/acrylic – AS/GC; no leaks noted

SALT/RAW WATER SYSTEMS: (CONT.)

Pump:	ShurFlo (m# 2095-212-834), 12V diaphragm – GC, FP
Discharge:	Live well, overboard aft, through discharge pump – ASWA; handle stiff
Hoses/Fittings:	Neoprene/to reinforced plastic, double clamps, nylon fittings – ASWA, FP
Light:*	Yes – AS, DNF ; (see Electrical)
Discharge Pump:	SeaLand TW12VDC, macerator pump; live well/deck box – ASWA, FP
Hoses/Fittings:	Neoprene, double clamps, bronze discharge seacock - AS; handle stiff
Bilge Pumps:	Four ; one Rule 1500gph/auto switch, forward engine room – GC, FP one Rule 1500gph, auto switch, aft engine room; auto only – FP one Rule 1500gph, helm/auto switch, aft engine room – FP one Rule 1500gph, helm/auto switch, aft/steerage – FP
Auto-Switches:	Four , Ultra PumpSwitch Mini – FP, all ‘on’ lights FP
Hoses/Fittings:	Reinforced plastic, single clamps – ASWA
Valves:	Inline check valves; Marelon hull seacocks – all AS; handles FP

COMMENTS AND OBSERVATIONS:

Salt water inlet, bronze seacocks include two engines, one generator, one wash down, one live well, and one A/C cooling, each with hull scoops, and canister strainers. Discharges include two engine and genset exhausts, the head discharge seacock, and the A/C outflow. Bilge pumps with automatic switches are located forward and aft in the engine room, and in the aft steerage.

NOTE: One bilge pump has been removed from the aft bilge, with the discharge components still intact. Replace the aft bilge pump, and prove the pump system as needed. The salt wash down system appears sound, however the outlet hose bib handle is frozen in place. Service the wash down hose bib, and prove the wash down system as needed.

E. FUEL SYSTEM:

Storage Tank(s):	Two
Fuel Type:	Gasoline
Construction:	5052 aluminum, welded box, painted gray – ASWA
Capacity:	276 US gallons total ; (each labeled 138 gallons)
Location:	Port/starboard outboard in engine room
Accessibility:	Limited; not readily accessible
Secured:	FRP tabbing/fasteners to bulkheads/stringers
Grounding:	Yes; wiring noted at each tank – ASWA; terminals secure
Labeled:	Yes; legible – GC
Level Indicator(s):	Two helm analog gauges – starboard FP; port DNF ; (see Electronics)
Tank Ventilation:	Conventional, overboard; port/starboard hull sides – ASWA; not proved
Fuel Transfer:	N/A
Filter Systems:	Two Mercury, primary water separators, engines – AS/GC; (see Engines) Sierra (p# 18-7844), primary water separator, genset – AS/GC secondary canister at genset; (see Generator)
Fill Pipe(s):	Two stainless, starboard cockpit cap, labeled ‘Gas’ – GC
Caps/Seals:	Two stainless, O-ring seals; chain tethers – GC overall
Grounding:	Not readily accessible; secured panel access
Hoses/Fittings:	Neoprene A2, double clamps at tanks – ASWA; fill pipes not accessible

FUEL SYSTEM: (CONT.)

Supply Lines/Hoses: Neoprene A1/single clamps; Aeroquip, bronze ends/fittings – ASWA
Fill Overflow: Overboard, through tank vent fittings – ASWA; not tested
Shut-Off Valves: None sighted
Priming Pumps: Three Parker (cat# V5C72120), inline 12V; (see **Engines/Genset**)
Carburetor Arrestors: Yes – GC; (see **Engines**)

COMMENTS AND OBSERVATIONS:

The vessel's gasoline fuel system appears sound and functions properly underway in sea-trial conditions. The fuel tanks are not readily accessible, and therefore their overall condition cannot be fully determined or guaranteed. No gasoline leaks or fumes were noted at the time of survey.

F. PORT PROPULSION SYSTEM:

Type: MerCruiser four-stroke inboard gasoline engine; direct drive
Model #: MIE8.1S Horizon
Serial #: Unknown; tag not legible/no numbers
Year Manufactured: Unknown – reported original equipment
Hour Meter: 582.0 at helm gauge – AS/FP; (reported accurate)
Configuration: Vee, eight-cylinder
Horsepower: 370HP @ 4,200 RPMs (reported online)
Propeller: Nibral, four-blade left hand (# 18 L 22), L-Cup – GC; true within 1/16"
Shaft: Stainless steel, 1-1/2" dia., tapered, locknut/cotter pin – ASWA
Packing: Tides Marine dripless, water-cooled; neoprene sleeve; spare ring – ASWA
Coupler: Four-bolt, lock washers, safety bolt/cable – ASWA/FC; **mild surface rust**
Struts: Single bronze 'I'-strut, 6-bolt, bonded – ASWA/GC overall
Cutless Bearings:* Primary/conventional – AS; **mild play noted**
Anodes: **Two**, fixed to shaft – AS/GC; ~ 85% left
Cooling System: Raw water cooled; closed coolant heat exchanger – ASWA
Inlet: Bronze seacock – AS/GC; handle stiff
Strainer: Perko (m# 493-500), bronze/acrylic, stainless basket – GC; no leaks
Hoses/Fittings: Neoprene, single/double clamps – ASWA; GC
Mounts: Single post vibration isolation, through-bolts to stringer 'T's – ASWA
Alternator: Belt drive, 12V/65A (reported) – ASWA; ~ 14.4V FP at helm gauge
Belts: Single serpentine, with tensioner – ASWA, **no shroud cover**
Fluid Levels: Lube oil level **satisfactory**; coolant **adequate**; no overflow reservoir
Lube Filter: Mercury (p# 35-866340K01), spin-on canister – AS/external GC, no leaks
Fuel Filter: Mercury (p# 35-802863T) primary water separator – AS/external GC
Air Filter: MerCruiser, pleated box element – ASWA
Carb Arrestor: Yes, fitted at carburetor forward – AS/GC; secure
Exhaust System: Raw water cooled, insulated cast iron risers, Centek FRP muffler, neoprene hose connections, all double clamps; port aft discharge – ASWA
Transmission: ZF Marine; marine reduction gear
Model #: HSW 630 A1
Serial #: 29684 K
Ratio: 2.04 : 1
Gear Oil/Filter: Oil level **adequate/appears clean**; no filter found

PORT PROPULSION SYSTEM: (CONT.)

Engine Shifter/Throttle Controls:	TeleFlex push/pull cable, two port-lever helm – AS/FP underway
Wiring:	Factory harness, properly routed, secured and terminated where accessible
Circuit Protection:	50A reset; thermal/hydraulic at engine panel – AS, not proved
Priming Pump:	Parker remote 12V pump, forward at engine space – AS/FC; not proved
Bilge Blower:	Attwood Turbo4000 12V 4”, port aft, ducting to vent insert – FP
Instruments:	Sea Ray analog gauges at helm – power up, AS/GC; (see Electronics)
Alarms:	Sea Ray ‘low oil’ audible at helm – FP; overheat reported/not tested
External Condition:	Good overall condition; mild surface corrosion; coupler/brackets/fasteners
Corrosion Protection:	Cooling system pencil zinc anodes – AS; monitor routinely

COMMENTS AND OBSERVATIONS:

The port engine and its external components and running gear appear in good overall condition, except as noted below. This vessel was tested in basic sea-trial conditions; see the Performance Tables at the end of this report. The market value stated in this report assumes the port engine will need only routine service and maintenance, and not major repairs.

NOTE: There is mild movement between the port drive shaft and the Cutless bearing. Service the port Cutless bearing at the next major haul-out as recommended by a reputable technician.

G. STARBOARD PROPULSION SYSTEM:

Type:	MerCruiser four-stroke inboard gasoline engine; direct drive
Model #:	MIE8.1S Horizon
Serial #:	Unknown; tag not legible/no numbers
Year Manufactured:	Unknown – reported original equipment
Hour Meter:	739.3 at helm gauge – AS/FP; (reported accurate)
Configuration:	Vee, eight-cylinder
Horsepower:	370HP @ 4,200 RPMs (reported online)
Propeller:	Nibral, four-blade right hand (# 18 R 22), L-Cup – GC; true within 1/16”
Shaft:	Stainless steel, 1-½” dia., tapered, locknut/cotter pin – ASWA
Packing:	Tides Marine dripless, water-cooled; neoprene sleeve; spare ring – ASWA
Coupler:	Four-bolt, lock washers, safety bolt/cable – ASWA/FC; mild surface rust
Struts:	Single bronze ‘I’-strut, 6-bolt, bonded – ASWA/GC overall
Cutless Bearings:*	Primary/conventional – AS; mild play noted
Anodes:	Two , fixed to shaft – AS/GC; ~ 90% left
Cooling System:	Raw water cooled; closed coolant heat exchanger – ASWA
Inlet:	Bronze seacock – AS/GC; handle stiff
Strainer:	Perko (m# 493-500), bronze/acrylic, stainless basket – GC; no leaks
Hoses/Fittings:	Neoprene, single/double clamps – ASWA; GC
Mounts:	Single post vibration isolation, through-bolts to stringer ‘T’s – ASWA
Alternator:	Belt drive, 12V/65A (reported) – ASWA; ~ 14.0V FP at helm gauge
Belts:	Single serpentine, with tensioner – ASWA, no shroud cover
Fluid Levels:	Lube oil level satisfactory ; coolant satisfactory ; no overflow reservoir
Lube Filter:	Mercury (p# 35-866340K01), spin-on canister – AS/external GC, no leaks
Fuel Filter:	Mercury (p# 35-802863T) primary water separator – AS/external GC
Air Filter:	MerCruiser, pleated box element – ASWA

STARBOARD PROPULSION SYSTEM: (CONT.)

Carb Arrestor: Yes, fitted at carburetor forward – AS/GC; secure
Exhaust System: Raw water cooled, insulated cast iron risers, Centek muffler, neoprene hose connections, all double clamps; starboard aft discharge – ASWA
Transmission: ZF Marine; marine reduction gear
Model #: HSW 630 A1 (match to port engine)
Serial #: Unknown; number plate missing
Ratio: 2.04 : 1 (match to port engine)
Gear Oil/Filter: Oil level **adequate/appears clean**; no filter found
Engine Shifter/ Throttle Controls: TeleFlex push/pull cable, **two** starboard-lever helm – AS/FP underway
Wiring: Factory harness, properly routed, secured and terminated where accessible
Circuit Protection: 50A reset; thermal/hydraulic at engine panel – AS, not proved
Priming Pump: Parker remote 12V pump, forward at engine space – AS/FC; not proved
Bilge Blower: Attwood Turbo4000 12V 4”, starboard aft, duct to vent insert – FP;
* **ducting from bilge not connected**
Instruments: Sea Ray analog gauges at helm – power up, AS/GC; (see **Electronics**)
Alarms: Sea Ray ‘low oil’ audible at helm – FP; overheat reported/not tested
External Condition: Good overall condition; mild surface corrosion; coupler/brackets/fasteners
Corrosion Protection: Cooling system pencil zinc anodes – AS; monitor routinely

COMMENTS AND OBSERVATIONS:

The starboard engine and its external components and running gear appear in good overall condition, except as noted below. This vessel was tested in basic sea-trial conditions; see the Performance Tables at the end of this report. The market value stated in this report assumes the starboard engine will need only routine service and maintenance, and not major repairs.

NOTE: The starboard bilge blower ducting from the bilge is not connected to the blower. Re-connect the starboard bilge blower inlet ducting as necessary to properly vent the engine space. There is mild movement between the starboard drive shaft and the Cutless bearing. Service the starboard Cutless bearing at the next major haul-out as recommended by a reputable technician.

H. GENERATOR:

Type:* Westerbeke, 3-cylinder, 1,800 RPM diesel engine – **DNFP/not run**
Model #: 4.5BCGB
Serial #: Genset – 45291; Engine – 029603-E106
Fuel: Gasoline
Output: 4.5kW; 38/19 Amps at 120/240V Volts (labeled)
Voltage: 120/240V at 60Hz; 12V start
Location: Starboard outboard in engine room
Mounts: Single post vibration isolation, mounted linear – ASWA
Start/Stop: At cabin/salon AC main panels – AS; **DNF/starter**
Hour Meter: 261.7 at genset panel
Fuel Lines/Hoses: Aeroquip/swaged brass ends; to copper tubing manifold – ASWA
Fuel Filter: Sierra (p# 18-7843), spin-on – AS; (see **Fuel System**)
secondary inline canister at genset inlet – not sighted
Oil Filter: Westerbeke (p# 39618) spin-on canister – AS/external GC; no leaks

GENERATOR: (CONT.)

Cooling System:	Raw water, through closed coolant heat exchanger – AS, no leaks noted
Inlet:*	Bronze seacock – AS; handle stiff/not proved
Strainer:	Perko bronze/acrylic, stainless basket – AS/GC; clean routinely
Siphon Break:	N/A
Discharge:	Through exhaust – ASWA; not proved/not in use
Hoses/Fittings:	Neoprene, double clamps – inlet hose in poor condition
Exhaust System:	Raw water cooled, starboard quarter aft discharge – ASWA; not run
Muffler:	Centek, molded FRP water-lift – AS/GC
Hoses/Fittings:	Neoprene, double clamps – ASWA
Fluid Levels:	Lube oil adequate ; coolant satisfactory ; overflow empty ; add per label
Belts:	Two conventional V; water pump/alternator – ASWA; tension good
Alternator:	Belt drive 12V – ASWA; not proved
Wiring:	Marine grade harness, properly routed, secured, terminated where sighted
Insulation:	N/A
Drain Pan:	Molded tray below unit – AS; some water/oily residue
Corrosion Protection:	Pencil anode in heat exchanger – monitor often

COMMENTS AND OBSERVATIONS:

The vessel's gasoline generator appears in good overall condition, except as noted below, and was not run at the time of inspection, due to reported starting issues. The market value stated in this report assumes the generator **will need only routine parts and service** to return the unit to proper function. The owner reports the generator will run properly by sale closing.

NOTE: The generator cooling inlet seacock handle does not move with moderate force, and was not proved, and the neoprene hose, from the seacock to the strainer, is in poor condition, aged and cracked throughout. Exercise and prove the generator cooling inlet valve, and the replace the cooling inlet neoprene hose as necessary.

I. AIR CONDITIONING SYSTEMS:

Number/Type:	One 115V reverse cycle, cool/heat zone unit
Model:	Dometic (m# DTU16-410A) – GC, FP
Capacities (BTUs):	16,000
Handler:	Integrated, 115V fan-coil unit – AS; clean intake filter routinely
Control Relay:	N/A; single unit
Cooling System:	Raw water cooled, copper tubing heat exchangers
Inlet:	Bronze seacock – AS, handle FP, no leaks
Strainer:	Perko, bronze/acrylic – AS/GC; no leaks
Pump:	Dometic (m# LC-3CP-MD), 115V vane pump – GC, FP
Discharge:	Starboard aft; above waterline – ASWA; good flow
Hoses/Fittings:	Neoprene, single/double clamps – ASWA; no leaks
Condensates:	To Sump Box ; salon bilge – ASWA; (see Sanitation)
Control Units:	One Dometic digital panel at salon – GC, cool/heat FP
External Condition:	Good overall condition
Drain Pans:	Factory; ABS plastic – GC; clean

AIR CONDITIONING SYSTEMS: (CONT.)

COMMENTS AND OBSERVATIONS:

The vessel has one air conditioner compressor/condenser zone unit, with integrated air handler, and three vents in the cabin. Cool functions properly, and heat powers up, and appears to function, but was not proved at the time of inspection.

J. ELECTRICAL SYSTEMS:

<u>VOLTAGE AC:</u>	120V; two 30A/125V shore, generator; engine alternators
Shore Inlet(s):	Two Marinco 30A/125V, covered at port cockpit – GC, FP
Shore Power Cable:	Two; one Marinco 30A/125V, yellow 25' – F/GC, FP; mild oxidation one Eel 30A/125V, yellow 25' – GC, FP one Eel 50A/dual 30A splitter/adaptor pigtail – GC, FP at survey
Main Breaker(s):	Two Marinco 30A main AC toggles cockpit locker – AS/GC; on/off FP
AC Panels:	Main salon switch panel; 7-120V toggles, labeled – GC; on/off FP two -line shore/genset slide isolators – AS/GC; genset not proved
AC Gauges:	Analog; two Amps/ two Volts – power up, AS/GC reverse polarity lights; dual 'power on' lights – AS/GC; not proved
AC Wiring:	Factory marine grade stranded copper/thermo-plastic harness, properly routed, secured, and terminated where accessible
Galvanic Isolation:	Two Professional Mariner ZincSaverII, 30A units – AS/GC; not proved
AC Outlets:*	Throughout the vessel, GFCI at galley/engine room – starboard FP/test wired OK; port outlets DNF at inspection
AC Lighting:	N/A
AC Appliances:	(See Galley/Additional Equipment sections)
AC Inverter:	N/A
<u>VOLTAGE DC:</u>	12V; five ship's batteries
Batteries:	Four West Marine (p# 15020234; g# 24), 12V lead acid – AS/GC; 5/19 one West Marine (p# 15020191; g# 27), 12V lead acid, genset – AS/GC
Secured:	Metal/threaded brackets to fixed trays – ASWA; mostly secure
Terminals:	Neoprene covers at (+) terminals – GC/secure
Fasteners:	Conventional fasteners – ASWA; terminals secure
Location:	Forward engine room, port/starboard/genset
Main DC Switches:	Three Guest rotary on/off, engines/genset/house – GC; on/off FP two main engine master key switches at helm – GC, FP
Battery Charger:	ProMar ProSport20Plus 12V/20Amp; 3-bank – powers up, GC/FP
Battery Isolator:	None sighted
DC Wiring:	Factory marine grade stranded copper/thermo-plastic harness, properly routed, secured, and terminated where accessible
Circuit Protection:	Thermal/hydraulic breakers at main/distribution switch panels – AS/GC breakers at engine room panel; 2-50A (engines); 3-50A mains – AS/GC pop-out type, labeled below helm – AS/GC; not proved main 120Amp windlass breaker at helm – AS; not proved helm fuse block; inline fuses at some electronics – ASWA
DC Ground:	12V negative
DC Outlets:	One 12V socket type at helm – GC, tests 12V; mild corrosion inside

ELECTRICAL SYSTEMS: (CONT.)

DC Panels:	Salon main, 8-12V toggles, labeled/lit – AS/GC; on/off FP where tested
DC Gauges:	Analog; one Volts/ one Amps at salon panel; selector toggle – AS/GC Genset remote start/stop at salon panel – GC, FP
Cabin Lights:	Headliner, step courtesy, valance, décor, lockers; some LED – all FP
Cockpit Lights:*	Two overhead at radar arch; below deck/gunwale courtesy; live well light – GC, most FP; except port gunwale light/live well light DNF
Deck Lights:	N/A
Engine Room Lights:	Two halogen, forward bulkhead – GC, FP
Navigation Lights:*	Forward toe-rail running; stern; steaming at spar – GC, FP; except spar
Anchor Light:*	Two-part 360° insert spar at arch frame spar – AS; DNF; hard to access
Searchlight:*	GoLight, bow mount, helm remote panel – AS/GC, DNF
Underwater Lights:*	Port/starboard transom – FP; port fixture adrift; starboard bulbs out
Windshield Wipers:	Two stainless two arm – F/GC, FP; no washers; blades stiff
Windlass:	Lofrans Progress I, 12V helm/deck switches – GC; (see Ground Tackle)
Horn:	Dual tone 12V, starboard bow – GC, FP; shrill
Fans / Blowers:	Two exhaust fans; head/galley – GC, FP two Attwood 4”, 12V exhaust blowers – GC, FP; (see Starboard Engine)
12V/Hydraulic:	Two Warner 12V/hydraulic deck actuators, ridge deck lift – F/GC, FP Windshield vent 12V/hydraulic actuator – GC, FP

COMMENTS AND OBSERVATIONS:

The vessel's AC shore power and ship's battery DC electrical systems appear sound and function properly, except as noted below. The AC shore power was tested with a 50A/250V shore power, split to two 30A/125V shore inlets, using the vessel's cables and splitter adapter.

NOTE: The port transom underwater light functions, but the light fixture is not secured in place, allowing water to trickle into the vessel through the wiring conduit. Properly re-secure the port underwater light fixture in place as necessary. At the time of inspection, the two-part anchor and steaming navigation light spar insert does not function. Service and prove the anchor/steaming light as necessary. At the time of inspection, the port side 110V outlets do not function, and no GFCI outlet was found tripped. Service and/or prove the port side 110V outlets as needed. The remote searchlight system appears sound, but does not function. Service the searchlight control system, and prove the searchlight as needed. The port gunwale courtesy, and live well lights do not function. Service these two lights as needed.

K. ELECTRONICS / NAVIGATION:

Helm:	Two master key ignition; two ignition, two start/stop toggles – GC, FP Sea Ray engine monitor diagram display at helm – powers up, AS/GC
Instrumentation:*	Eleven Sea Ray analog; two RPMs/engine hours; oil pressure/ /Volts, coolant temps/fuel levels; one synchronizer – AS/GC; except port fuel
VHF Radio:	Standard Horizon Matrix (m# GX2000) – AS/GC; radio check FP
Speaker:	West Marine 4” remote at dash; hailer at arch – GC, dash speaker FP
Compass:	Ritchie PowerDamp, 4” white/white card, dash mount – GC/AS underway
GPS/Multi-Function Displays (MFDs):	One Garmin GPSMap7610xsv, 10” MFD, flush mount – GC; GPS, charts, radar, sonar, info, combinations, data FP; no SD/sim cards

ELECTRONICS / NAVIGATION: (CONT.)

Depth Sounder:	Garmin 7610xsvSonar, sounder processor – FP through MFD
Radar:	Garmin xHD2 (m# GMR624), 42” open scanner, 72 mile – AS/FP at MFD Garmin DC-DC radar power converter – ASWA
Auto Pilot:	Raymarine AutoHelm system: ST6002 AutoHelm control/display panel; SmartPilot processor; Type 1 12V/hydraulic steering pump (reported), auto-compass – system powers up, heading/steer FP underway
Transducers:	Three two Airmar removable through-hull; one speed log with plug; one depth, with plug – ASWA; not proved; no # tags found one Garmin through-hull fairing block – AS/GC, FP at MFD; no # tag
Antennas:*	One Shakespeare 8’ VHF, stainless ratchet mount – AS/GC; cable cracks one Raymarine Raystar120 remote GPS, short spar at arch – AS/GC
Phone/Cable TV:	Marinco phone/cable TV covered inlet – AS/GC, not tested
Satellite TV:	KVH TracVision M3, satellite TV dome receiver – AS/GC; not proved KVH control/interface box, salon locker – AS/GC; not proved DirecTV HD control box – AS/GC; new in packing; not installed
Entertainment:	Two TVs, satellite TV, media player, stereo system; TV/stereo select
Salon:	Panasonic OmniVision14” fixed TV/VHS player – AS/GC; not proved Radio Shack TV/media selector interface module – AS; not proved
Media:	Philips (m# DVP1120), remote DVD/disc player – AS/GC; not proved
Speakers:	Two overhead 7” flush – ASWA, FP with stereo
Cabin:	Samsung (m # LN.T1954H), bulkhead pivot mount – powers up, AS/GC
Speakers:	Two Clarion 6.5” flush outboard – ASWA, FP with stereo
Stereo:	Fusion (m# MS-AV650), radio/XM/CD/BT – GC, radio FP
Amplifier(s):	One Clarion (m# APX400.4M) – AS/GC; not proved one Marine Audio (m# MAA120-2) – AS/GC; not proved
Remote:*	Remote panel at helm dash – AS/GC; DNF/not in use
Auxiliary:	Dual USB charging port – GC, blue light on; not proved
Speakers:*	Five; four Kenwood 7”, cockpit coamings fore/aft – AS; three DNF one Kenwood 10” subwoofer; starboard salon overhead – AS; not proved
Additional:	Bennett trim tab toggle control/display panel – FP; (see Steering section) GoLight remote searchlight panel – AS/ DNF ; (see Electrical) GloMex TV antenna tuner dial panel – AS; no TV antenna found

COMMENTS AND OBSERVATIONS:

The vessel’s electronic navigation and communication systems power up normally and appear to function properly at the time of inspection, except as noted below. These systems were tested in sea-trial conditions. The electronic entertainment system components, with shore and satellite TV systems, and phone inlet appear sound, but were not proved.

NOTE: The port fuel level gauge does not function. Service and prove the port fuel level gauge as needed. The two port and starboard aft cockpit stereo speakers do not function at the time of inspection. Service and prove the cockpit stereo speakers where needed. The VHF antenna cable insulation has cracks near the base. Service or replace the VHF antenna cable, and prove the VHF radio transmits as necessary when installed.

L. HULL, DECKS, and SUPERSTRUCTURE:

Hull:	Molded FRP laminate, deep-Vee , hard chines; partial inner/mid/outer strakes; molded sheer-line spray-rails, drive tunnels – GC overall; no delamination; * two osmotic blisters noted
Bottom Coating:	Black anti-foul paint over barrier coat – GC overall
Keel/Ballast:	N/A / fuel, water, equipment ballast
Through-Hulls:*	Bronze seacocks; hull scoops; drain below the waterline – GC overall Marelon seacocks/chrome vent fittings above – AS; except two port aft
Bonding/Anodes:	Transom sacrificial anode block; green wiring – GC; continuity noted throughout running gear, aft through-hull fittings/drain; ~ 85% left
Grounding Plate:	Small Dyna-Plate, starboard hull aft – FC; not in use
Sides:	White gel coat, black boot top/gold accent stripes, molded arch with plastic vent fittings – GC overall; mild cosmetic gel coat oxidation
Transom:	Molded FRP/core, white gel coat/gated port door, red/black/yellow graphics/lettering – GC overall; mild oxidation
Swim Platform:	Molded FRP/white gel coat, fixed to transom, textured non-skid, three-step stainless fold-away boarding ladder/hatch – GC/secure; ladder FP
Decks:	White gel coat; textured non-skid overlay in all traffic areas – GC overall
Chain Locker:	Starboard deck hatch, white gel coat, drains port/starboard – ASWA/GC
Deck Hatches:	One Bomar tinted acrylic/aluminum – GC; seal/latches/hinge, pull-out screen/shade – GC overall, FP
Cockpit Hatches:	Molded FRP/core, texture non-skid, stainless hardware – GC
Deck Hardware:	Stainless pull-up/deck cleats, tank fills, rails – GC; secure
Hand-rail:	Stainless/stainless stanchions, fixed to molded rails aft 70% – GC; secure
Grab-rails:	Stainless; companionways, cabin, bridge, cockpit, transom – GC/secure
Hull-to-Deck Joint:	Overlap; fasteners/sealant secured – ASWA at bow locker
Rub-rail:	White PVC/stainless cap, full-perimeter – GC
Superstructure:	Integral FRP/core, white gel coat; black/gold cabin stripes/logos – GC
Portholes:	Two port/two starboard in cabin, stainless/glass – GC; no screens found
Bulkheads:	Laminate FRP/core, gray gel coat over where readily accessible – FC; some gel coat loose/separated; bare laminate exposed in the bilges FRP tabbing reinforcement at hull/intercostals/stringers – ASWA
Longitudinal:	Molded, heavy FRP laminate, FRP tabbing to floors/bulkheads – ASWA

COMMENTS AND OBSERVATIONS:

The hull was percussion sounded with a phenolic hammer approximately every six inches along and below the waterline where accessible while hanging in TravelLift slings, with no sign of delamination found, and only minor blistering, as noted below. The decks were tested for moisture and delamination in higher traffic areas, with no significant results for the vessel's age and location. The exterior gel coat and bright-work, as well as the interior finish work overall, appear in average/good overall condition for the vessel's age, except as noted below.

NOTE: The two port aft transom through-hull fittings are not well secured or sealed to the hull. Properly re-bed and secure the port aft through-hull fittings as needed. There is excessive debris build-up in the aft bilge. Clean the aft bilge of debris as needed. The sliding cabin door does not move easily. Adjust the sliding cabin door properly as needed. At the time of inspection, there is water build-up outboard of the engine outer stringers port and starboard in the engine space.

HULL, DECKS, and SUPERSTRUCTURE: (CONT.)

Recommend the trapped water be removed from the engine space where needed, and determine how to prevent water intrusion into these areas in the future. **Cosmetic findings** include: the topside gel coat is oxidized and in need of compounding and wax; the external cockpit and bridge seating upholstery is oxidized, with some stitching separation and tearing in some areas; there are two 4-5" osmotic blisters (see **Images**). Service these cosmetic findings where needed.

M. STEERING SYSTEMS:

Type:	Single hydraulic helm; twin spade rudder; single actuator/torsion bar
Helm:	SeaStar Classic Tilt 1.7 (p# HH6541), hydraulic – GC, FP underway
Wheel:	Sea Ray; three-spoke Euro style; hardwood rim – FC/FP; warped
Fluid Reservoir:*	Cap at SeaStar helm – AS/GC; no leaks noted; fluid level looks low
Rudders:	Two, bronze spade – GC
Stocks:	Stainless 1-½"; key ways/single clamps – ASWA
Packing:	Conventional, bronze glands/lock nuts – AS; FC; leak water/corrosion
Tables:	Two fixed bronze plate, aft stringers – F/GC; mild corrosion
Bonding:	Wiring noted at quadrant fasteners – ASWA; continuity noted
Stops:	N/A
Quadrants:	Two bronze tiller arm; single clamps/keyways/pins – ASWA
Torsion Bar:	Stainless steel, adjustable tangs ends – GC; mild surface rust
Hydraulic Assist:	SeaStar (m# BA135-7ATM; p# HC5313) aluminum cylinder, chrome ram actuator; fixed at starboard table – GC, FP; secure/no leaks noted
Steering Pump:	N/A; (see Auto-Pilot)
Steering Lines/Hoses:	Reinforced plastic, compression ends/fittings – ASWA, no leaks noted
Auto Pilot:	Raymarine; 12V/hydraulic pump not readily accessible; (see Electronics)
Trim Tabs:	Bennett stainless plate, single ram, transom mount w/ anti-foul paint; Bennett V-351 hydraulic pump/reservoir; dual toggle helm panel – all GC; FP underway; fluid level satisfactory at reservoir; tab anodes < 50%
Bow Thruster:	N/A
Emergency Tiller:	N/A
Additional:	Cetec Benmar H Power Unit, auto-pilot pump – AS; not in use/obsolete

COMMENTS AND OBSERVATIONS:

The vessel's single helm hydraulic steering system and components, with 12V/hydraulic trim tab system, and 12V/hydraulic auto-pilot system, appear sound and in good condition where accessible. The vessel steering systems were tested in sea-trial conditions. The auto-pilot system and components are covered in the **Electronics/Navigation section**.

NOTE: There is water trickling in at both the rudder packing glands, with moderate surface corrosion and salt build-up found. Service the two rudder packing glands for water intrusion and surface corrosion as needed. The steering fluid level appears low at the helm cap. Properly top-off the steering fluid to the level recommended by the manufacturer, and monitor fluid level routinely. The steering wheel appears slightly warped. Service or replace the steering wheel if needed.

N. GROUND TACKLE:

Anchor Platform:*	Integral platform, stainless deck flume, two rollers, StarBoard® deck pad, LED deck light – GC, light FP; flume bent, moderate to port
Chain Locker:	Starboard forward deck hatch, drains overboard port/starboard, with fresh water hose bib – ASWA/GC; (see Fresh Water)
Primary Anchor:	Danforth # 35-H, galvanized sand, at bow – GC
Chain:*	~ 25' galvanized 5/16" – ASWA/GC;
Secured:	Stainless swivel shackle to anchor; seized secure – GC
Rode:	~ 100' 9/16" three-strand, spliced to chain – not all removed from locker mild chain link rust at splice
Secured:	Tied to locker eye – AS/secure
Windlass:	Lofrans Progress I, 12V aluminum chain gypsy – GC, FP at the dock
Switches:	Helm control toggle; on/off toggle; two deck switches – GC, FP
Manual:	Lofrans handle, fixed in bow locker – AS/GC; not proved
Deck Hardware:	Three stainless deck cleats; stainless safety cable/clip – GC
Additional:	None found

COMMENTS AND OBSERVATIONS:

The vessel's primary anchor and ground tackle system is in good condition, and appears sound and sufficient for the vessel's intended day use. Recommend owner/operator remove ground tackle from chain locker **periodically**, inspect for condition, and confirm proper locker drainage. NOTE: The anchor platform flume is bent moderately to port, however appears to still function. Service the anchor flume cosmetically as needed.

O. SAFETY EQUIPMENT:

Life Jackets (PFDs):	Three Adult Type I , reflect tape – AS/GC; keep accessible seven Adult Type II ; six Adult/ five Child Type III – GC overall
Life Ring/Cushion:	One Jim-Buoy GW-20" life ring – GC; keep accessible; add poly line three Type IV cushions – GC; keep accessible
Fire Extinguishers:	Two ; one 10B:C, at bridge deck, gauge in green – AS/GC; not tagged one 5B:C, galley, gauge in green – AS/GC; not tagged
Fixed Fire System:*	Sea-Fire 1301 Halon (m# FH), manual pull cable/engine shut-down, panel, labeled – AS/GC; gauge in green; not tagged
Distress Flares:*	Four red flare handheld; kit with seven handheld; four 12-gauge/gun; distress flag – all AS/GC; all expired 2016/2018
First Aid:	Basic Medi-Kit; additional supplies noted – GC; AS for intended use
Horn/Whistle:	Handheld air-horn/new; 12V horn at bow – FP; (see Electrical)
'Trash' Placard:	Yes, properly displayed
'No Oil' Placard:	Yes, properly displayed
Bilge Pumps:	Four 12V centrifugal – GC, FP; (*see Salt/Raw Water)
High Water Alarm:*	Two hi-water auto switches; helm alarm horn – DNF ; switches not wired
E.P.I.R.B.:	N/A
Life Raft:	N/A
CO Fume Detectors:	Two Xintex CO detector units; cabin/galley – AS/GC; test FP
Smoke Alarms:	None found
Ship's Bell:	N/A

SAFETY EQUIPMENT: (CONT.)

USCG 'Rules': N/A

VHF Radio: Yes – GC, radio check FP; (see **Electronics**)

COMMENTS AND OBSERVATIONS:

The vessel's safety and USCG equipment is in good condition overall, and appears sufficient for the intended use, except as noted below. This equipment should be onboard at all times while underway, properly maintained, sufficient for each passenger, and readily accessible. Although the fixed fire suppression system appears sound, the NFPA **recommends** all marine fire suppression systems be inspected and tagged by a qualified technician at least every two years.

NOTE: At the time of inspection, the onboard distress flares appear in good condition, but all are past the expiration dates. Add additional distress flares as necessary to comply with USCG requirements **before next underway**. The fixed fire suppression system appears sound, however is not tagged. The NFPA recommends marine fire suppression systems be inspected and tagged by a qualified technician at least every two years. There are two Rule auto high bilge-water alarm switches, however they are not wired to an alarm system, and do not function. Service and prove the audible high bilge-water alarm system as necessary.

P. ADDITIONAL EQUIPMENT:

- + Molded 'L' shape removable cockpit bench seat module; vinyl upholstery; cockpit brackets, stored off vessel – AS, FC; upholstery oxidized; not set in place at survey
- + Beam 8 gallon/3.5HP, 110V central vacuum system; reinforced plastic hoses/nozzles – GC, FP
- + Tan canvas, helm, seat, cockpit console covers; zippers/snaps – F/GC; some oxidation
- + **Three** medium white /**one** small yellow dock fenders; **six** braided black dock lines – GC
- + **Two** stainless rod holder line-tenders; aluminum boat pole/brush – GC
- + Various spare Rule 1500gph pump, engine/genset filters, parts – all AS/GC; not inventoried
- + Vornado 110V circulation fan; 10" 110V circulation fan – both GC, FP
- + Owners/operators manuals, logs, vessel records, documentation, etc. – GC; not inventoried

IV. FINDINGS AND RECOMMENDATIONS:

Deficiencies will be listed under the appropriate heading:

A. SAFETY DEFICIENCIES

B. OTHER DEFICIENCIES NEEDING ATTENTION

C. Surveyor's Notes and Observations

Deficiencies noted under "**SAFETY DEFICIENCIES**" should be addressed before the vessel is next underway. These findings represent an endangerment to personnel and/or the vessel's safe and proper operating condition. ***Findings may also be in violation of USCG regulations.*** Deficiencies noted under "OTHER DEFICIENCIES" should be corrected in the near future to maintain standards and help retain the vessel's value.

A. GALLEY:

- B. The cockpit ice-maker appears sound, but does not function at the time of inspection, and the door latch is broken away – Service and prove the cockpit ice-maker, and replace the door latch as needed

B. SANITATION SYSTEMS:

- A. **The neoprene hose from the macerator pump to the below waterline discharge valve is aged, cracked throughout, and leaks when the discharge valve is opened**
 - Properly replace the head system discharge neoprene hose as necessary
- C. The neoprene hose from the vacuum tank to the holding tank is aged and cracking
 - Replace the holding tank inlet neoprene hose from the vacuum tank as needed

C. FRESH WATER SYSTEM:

- C. The fresh water wash down coiled hose in the bow locker is in poor condition
 - Replace the bow locker fresh water wash down hose as needed

D. SALT/RAW WATER SYSTEMS:

- B. One bilge pump has been removed from the aft bilge, with the discharge components still intact – Replace the aft bilge pump, and prove the pump system as needed
- B. The salt wash down system appears sound, however the outlet hose bib handle is frozen in place – Service the salt wash down hose bib, and prove the wash down system as needed

E. FUEL SYSTEM:

F. PORT PROPULSION SYSTEM:

- C. There is mild movement between the port drive shaft and the Cutless bearing – Service the port Cutless bearing at the next major haul-out as recommended by a reputable technician

G. STARBOARD PROPULSION SYSTEM:

- A. **The starboard bilge blower ducting from the bilge is not connected to the blower**
 - Re-connect the starboard bilge blower inlet ducting as necessary to properly vent the engine space

FINDINGS AND RECOMMENDATIONS: (CONT.)

- C. There is mild movement between the starboard drive shaft and the Cutless bearing
 - Service the starboard Cutless bearing at the next major haul-out as recommended by a reputable technician

H. GENERATOR:

- A. **The generator cooling inlet seacock handle does not move with moderate force, was not proved, and the neoprene hose, from the seacock to the strainer is in poor condition, aged and cracked throughout** – Exercise and prove the generator cooling inlet valve, and the replace the cooling inlet neoprene hose as necessary

I. AIR CONDITIONING SYSTEMS:

J. ELECTRICAL SYSTEMS:

- A. **The port transom underwater light functions, but the light fixture is not secured in place, allowing water to trickle into the vessel through the wiring conduit**
 - Properly re-secure the port underwater light fixture in place as necessary
- A. **At the time of inspection, the two-part anchor and steaming navigation light spar insert does not function** – Service and prove the anchor/steaming light as necessary
- B. At the time of inspection, the port side 110V outlets do not function, and no GFCI outlet was found tripped – Service and/or prove the port side 110V outlets as needed
- B. The remote searchlight system appears sound, but does not function – Service the remote searchlight control system, and prove the searchlight as needed
- C. The port gunwale courtesy light, and live well light do not function – Service these two lights as needed

K. ELECTRONICS / NAVIGATION:

- B. The port fuel tank level gauge does not function – Service and prove the port fuel tank level gauge system as needed
- B. The **two** port and **one** starboard aft cockpit stereo speakers do not function at the time of inspection – Service and prove the cockpit stereo speakers where needed
- C. The VHF antenna cable insulation has cracks near the base, exposing co-axial cable inside – Service or replace the VHF antenna cable, and prove the VHF radio transmits as necessary when installed

L. HULL, DECKS, and SUPERSTRUCTURE:

- B. The two port aft transom through-hull fittings are not well secured or sealed to the hull – Properly re-bed and secure the port aft through-hull fittings as needed
- B. There is excessive debris build-up in the aft bilge compartment – Clean the aft bilge and fixed components of debris as needed
- C. The sliding cabin companionway door does not move easily – Adjust the sliding cabin door as needed to open and close properly
- C. At the time of inspection, there is standing water build-up outboard of the engine bay outer stringers port and starboard in the engine space – Recommend the trapped water be removed from the engine space where needed, and determine how to prevent water intrusion into these areas in the future

FINDINGS AND RECOMMENDATIONS: (CONT.)

- C. **Cosmetic findings** include: the topside gel coat is oxidized and in need of external compounding and wax; the external cockpit and bridge seating upholstery is oxidized, with some stitching separation and tearing in some areas; there are **two** 5+” osmotic blisters below the waterline – Service these cosmetic findings where needed

M. STEERING SYSTEMS:

- B. There is water trickling in at both the rudder packing glands, with moderate surface corrosion and salt build-up found – Service the two rudder packing glands for water intrusion and surface corrosion as needed
- C. The steering fluid level appears low at the helm cap – Properly top-off the steering fluid level as recommended by the manufacturer, and monitor fluid level routinely
- C. The steering wheel appears slightly warped – Service or replace the steering wheel if needed

N. GROUND TACKLE:

- B. The anchor platform stainless flume is bent moderately to port, however appears to still function – Service the anchor flume cosmetically as needed

O. SAFETY EQUIPMENT:

- A. **At the time of inspection, the onboard distress flares appear in good condition, but all are past the printed expiration dates** – Add additional distress flares as necessary to comply with USCG requirements **before next underway**
- B. The fixed fire suppression system appears sound, however is not tagged – The NFPA recommends all marine fire suppression systems be inspected and tagged by a qualified technician at least every two years
- B. There are two Rule auto high bilge-water alarm switches, however they are not wired to an alarm system, and do not function – Service and prove the audible high bilge-water alarm system as necessary

P. ADDITIONAL EQUIPMENT:

V. SUMMARY AND VALUATION

A. STATEMENT OF VALUATION:

The “**Estimated Present Day Market Value**” is the most probable price in terms of money which a vessel should bring in a competitive and open market under all conditions requisite to a fair sale, the buyer and seller, each acting prudently, knowledgeably and assuming the price is not affected by undue stimulus. After reviewing values of the available Sea Ray 340 vessels currently for sale on several web-based sites; the vessel’s **Average** (BUC) overall condition, with twin gasoline inboard engines, a gasoline generator, some updated electronic navigation and entertainment equipment, comparable factory options, and additional equipment; as well as value ranges for this year and model vessel and engines from BUCValuPro.com (119th edition), it is your surveyor’s opinion that the “**Estimated Present Day Market Value**” of the subject vessel **at the time of survey** is approximately:

\$67,800.00

Sixty-Seven Thousand, Eight Hundred U.S. Dollars

The “**Estimated Replacement Cost**” indicates the retail cost of a new 2020 Sea Ray Sundancer 350 (the closest current model for 2020), with twin MerCruiser gasoline inboard engines, a generator, comparable electronics and factory options, and additional equipment, offered by a current dealer in South Florida. Sea Ray Boats online website ‘Build Your Sea Ray’, and BUCValuPro.com (119th edition) suggest an approximate “**Estimated Replacement Cost**” of the subject vessel comparably equipped and delivered as:

\$455,000.00

Four Hundred Fifty-Five Thousand U.S. Dollars

B. SUMMARY:

In accordance with the request for a marine survey of the M/V “**Black Jack**”, for the purpose of evaluating its present condition and estimating its Present Day Market Value and Replacement Cost, I herewith submit my conclusion based on the preceding report. The subject vessel was personally inspected by the undersigned on **July 27, 2020**, and was found to be in **Average** overall condition. The vessel was tested in sea-trial conditions. Subject to correction of deficiencies listed in **Section IV as A. (Safety Findings)**, the vessel is considered to be suitable for its intended use. Other deficiencies listed should be attended to in a timely fashion.

This report is submitted without prejudice and for the benefit of whom it may concern.



ATTENDING SURVEYOR:

Scott M. Austin AMS®

DATE: July 28, 2020

Member of the Society of Accredited Marine Surveyors, SAMS® AMS #1184

C. IMAGES SECTION:



M/V 'Black Jack' – Port bow forward



Starboard bow forward



Port quarter aft



Starboard quarter aft



Port side running gear



Starboard side running gear



Port MerCruiser MIE8.1S gasoline inboard engine



Starboard MerCruiser MIE8.1S gasoline inboard engine



Head system discharge neoprene hose/leaks



Holding tank inlet neoprene hose aged



Port aft outboard through-hull fitting not secure



Port aft inboard through-hull fitting not sealed



Osmotic hull blister starboard forward



Osmotic hull blister port forward



Port transom underwater light fixture not secure



Port underwater light in transom, leaks water in



VHF antenna cable insulation cracks



Aft bilge debris; foul water build-up



Anchor/steaming light spar insert outlet at arch



Genset cooling inlet neoprene hose aged/cracks

D. SEA-TRIAL PERFORMANCE TABLES:

RPMs from the Sea Ray analog helm gauges / SOG/MPH from the helm Garmin 7610xsv MFD

* No fuel management data found

Vessel run in 5-8 KTs East winds; seas to 1 foot; 2 Adults onboard; fuel near full; water near ½
Near-coastal, Bay side, Islamorada, FL

Upwind/Crosswind – no trim tabs:

PORT ENGINE:	Idle	Slow			Plane	Cruise				WOT (Downwind)
RPMs	750	1,000	1,500	2,000	2,500	3,000	3,400	3,800	4,050	4,100
SOG (MPH)	4.6	6.3	8.1	9.1	10.4	22.6	28.2	32.3	33.7	34.5
STARBOARD										
ENGINE:	Idle	Slow			Plane	Cruise				WOT (Downwind)
RPMs	750	1,000	1,500	2,000	2,500	3,000	3,400	3,800	4,150	4,200
SOG (MPH)	4.6	6.3	8.1	9.1	10.4	22.6	28.2	32.3	33.7	34.5

NOTE: The Downwind Wide Open Throttle (WOT) was performed while adjusting the trim tabs underway for maximum Speed Over Ground (SOG)